Leakage-Aware Brain Tumour MRI Classification with a Calibrated, Explainable EfficientNet-B0

A white paper on data hygiene, explanation quality and an in-browser research demo

Rishabh Aryan

M.Tech (Artificial Intelligence and Data Science), Department of Computer Science and Engineering
Indian Institute of Information Technology Bhagalpur, Bihar, India

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Abstract

Public brain MRI collections are widely used to benchmark four-way tumour classifiers (glioma, meningioma, pituitary tumour, no tumour), and reported accuracies are often above 94% [4, 8, 24]. We revisit the popular Kaggle *Brain Tumor MRI Dataset* (version 2, 7,200 images) with an emphasis on evaluation hygiene. A perceptual-hash audit shows that, although no file is byte-identical across the official Training and Testing folders, 441 of the 1,600 Testing images (27.6%) have a strict near-duplicate in Training, concentrated in the no-tumour class. We remove every Training image linked to a Testing image, split the remainder into train and validation sets by duplicate group, and fine-tune an ImageNet-pretrained EfficientNet-B0 on a CPU. On the full Testing folder the model reaches 91.9% accuracy (95% bootstrap CI 90.5%–93.2%), macro-F1 0.919 and macro one-vs-rest ROC-AUC 0.987, while validation accuracy inside the training distribution is 98.7%. An identical model trained on the original folders scores 96.1%, but it is right on 105/105 Testing meningiomas that have near-copies in Training against 55/105 for the leakage-aware model, suggesting that much of this gap reflects memorisation of near-copies. Temperature scaling ($T = 1.91$) reduces the expected calibration error on Testing from 0.070 to 0.052, and a confidence threshold chosen on validation flags 3.5% of Testing predictions as uncertain. We compare five attribution methods (Grad-CAM, Grad-CAM++, Integrated Gradients, occlusion and KernelSHAP on superpixels) with the model-parameter randomisation sanity check and deletion/insertion faithfulness metrics: Grad-CAM passes the sanity check and, with Grad-CAM++, has the highest insertion AUC, whereas Integrated Gradients fails the sanity check despite the best deletion score of the five methods, which only matches a random ordering. The model, exported to ONNX, runs in the browser with exact Grad-CAM, occlusion and a templated "why this prediction" panel that uses only model outputs and measured test statistics. The system is a research and education artefact, not a medical device.

1 Introduction

Brain tumours are a heterogeneous group of neoplasms whose diagnosis rests on imaging, histology and, since the 2021 WHO classification, molecular markers [13]. Magnetic resonance imaging is the main imaging modality, and automatic slice-level classification of MRI into broad tumour categories has become a standard exercise for transfer learning in medical imaging [4, 8, 24]. Much of this work uses small public collections in which several images come from the same patient or the same scan, and in which the provenance of individual images is only partly documented.

Two problems follow. First, when near-identical slices sit on both sides of a train/test split, test accuracy partly measures memorisation rather than generalisation, a form of leakage that has repeatedly inflated results in applied machine learning [11, 20]; even curated vision benchmarks contain such near-duplicates [5]. Second, the saliency maps that often accompany these classifiers are rarely tested: they can look plausible while being insensitive to the model's parameters [2], and in medical images they frequently fail to localise pathology [3, 21].

This white paper documents a small, fully reproducible project that tries to address both. Our contributions are:

1. an audit of exact and near-duplicate images in version 2 of the Kaggle Brain Tumor MRI Dataset [16], with all hashes, pairs and split files released for review;
2. a leakage-aware split and an EfficientNet-B0 [25] baseline trained on a CPU, reported with confidence intervals, per-class metrics, ROC-AUC, calibration and an explicit uncertainty threshold, and compared with a naive split of the same data;
3. a comparison of five attribution methods with sanity checks and faithfulness metrics; and
4. an in-browser demonstrator that reproduces the PyTorch probabilities, computes Grad-CAM exactly from ONNX outputs, and explains each prediction using only measured quantities.

2 Related work

Brain tumour classification. Cheng et al. [7] released the figshare T1-weighted contrast-enhanced collection (glioma, meningioma, pituitary tumour) and classified tumour regions with handcrafted features after region augmentation and partitioning. Deepak and Ameer [8] classified features from an ImageNet-pretrained GoogLeNet [9] with classical classifiers and Swati et al. [24] fine-tuned a pretrained VGG block-wise, both reaching high accuracy on the same data (98% and 94.82%), and Badža and Barjaktarović [4] reported a compact CNN evaluated with patient-wise and record-wise cross-validation, noting the gap between the two (96.56% record-wise). The Kaggle dataset used here merges figshare with the SARTAJ and Br35H collections, so these figures are not on the same test set as ours [16]. Volumetric multi-sequence benchmarks such as BraTS [15] target segmentation of gliomas rather than slice-level type classification.

Leakage and duplicates. Kapoor and Narayanan [11] catalogue leakage as a main cause of irreproducible machine-learning findings, and Roberts et al. [20] found that none of the 62 COVID-19 imaging models they reviewed was of potential clinical use, owing to methodological flaws and biases that included duplicated images across combined datasets. Barz and Denzler [5] purged near-duplicates from CIFAR and measured the resulting accuracy drop. We use perceptual hashing [27] combined with pixel correlation to find such pairs.

Explanations and their evaluation. Occlusion sensitivity [28], Grad-CAM [22] and Grad-CAM++ [6], Integrated Gradients [23], LIME [19] and SHAP [14] are common post-hoc attribution methods. Adebayo et al. [2] proposed the model-parameter randomisation test, and Petsiuk et al. [18] introduced the deletion and insertion metrics. Benchmarks in radiology report that saliency maps often fall short of localisation targets [3, 21]. Guo et al. [10] showed that modern networks are often over-confident and that temperature scaling corrects this effectively.

3 Data and duplicate audit

3.1 Dataset

Version 2 of the Brain Tumor MRI Dataset (13 February 2026, CC BY 4.0) holds 7,200 2-D slices in a `Training` folder (1,400 per class) and a `Testing` folder (400 per class). According to its description, meningioma and pituitary images come from figshare [7] and SARTAJ, glioma images from figshare (the SARTAJ gliomas were replaced in curation), and no-tumour images from Br35H; and version 2 removed duplicates and eliminated train/test overlap. Images vary in size (most are 512×512) and in sequence and orientation (axial, coronal and sagittal; T1, T1 with contrast, T2 and FLAIR); no patient identifiers are provided. We download it with `kagglehub` and never redistribute it, apart from eight resized sample slices in the demo, as the licence allows with attribution.

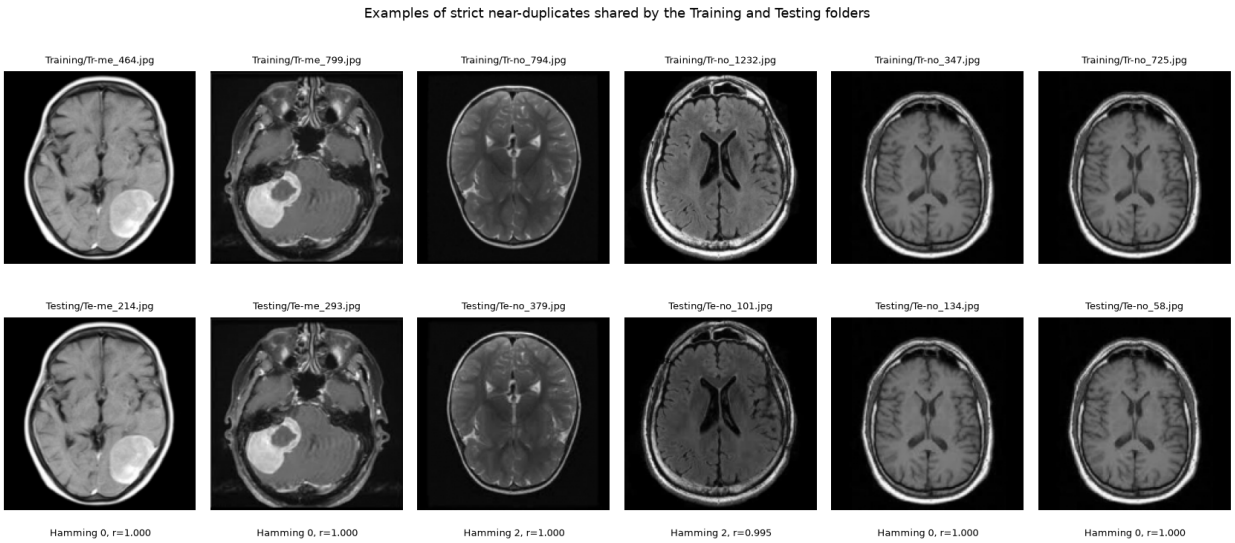


Figure 1: Strict near-duplicates shared by the Training and Testing folders (top: Training file, bottom: matching Testing file, with pHash Hamming distance and thumbnail correlation).

3.2 Audit method

For each image we compute the MD5 of the file, a 64-bit DCT perceptual hash (pHash, `imagehash`) of the greyscale image and a 64×64 zero-padded thumbnail. For every pair within a Hamming distance of 10 we compute the Pearson correlation of the thumbnails. Visual inspection of random pairs at each distance showed that pHash alone is too coarse for MRI (many different slices of similar anatomy sit 4–6 bits apart), so we define:

- **strict near-duplicate**: identical MD5, or Hamming distance ≤ 2 and correlation ≥ 0.98 (in our inspection, the same image re-encoded or resized);
- **linked**: identical MD5, or Hamming distance ≤ 6 and correlation ≥ 0.95 (also catches near-identical neighbouring slices of the same scan), or Hamming distance ≤ 10 and correlation ≥ 0.98 (copies whose contrast or sharpening was altered, which move the pHash by 8–10 bits).

Connected components of linked images form *groups*. The first two thresholds were fixed by inspection before any model was trained. The third clause was added after an internal review found contrast- or sharpening-altered copies of Testing images at Hamming distance 8–10 that the first version of the rule missed; it adds 215 linked pairs (71 across the Training/Testing boundary), raises the number of removed Training images from 1,198 to 1,223, and the model was retrained on the rebuilt split. The the full candidate-pair list (Hamming ≤ 10) is released so that others can apply different thresholds.

3.3 Findings

There are 340 files involved in exact duplication (7,013 unique MD5 values), none of them across folders, which agrees with the dataset description. Near-duplicates, however, are common: we find 3,484 strict pairs, 1,270 of them across the Training/Testing boundary, and 6,177 linked pairs (2,229 across). 441 Testing images (27.6%) have a strict near-duplicate in Training and 630 (39.4%) share a linked group with a Training image. The overlap is class-dependent: strict cross-split duplicates affect 327 no-tumour, 105 meningioma, 9 glioma and 0 pituitary Testing images. 13 images belong to groups that carry more than one class label. The official Testing folder itself contains near-identical images with different labels (for example `Te-gl_68` and `Te-me_96` at Hamming distance 0, correlation 0.9999), so a few test errors may be unavoidable label noise. Figure 1 shows examples. In addition, 100 Training and 103 Testing meningioma files are augmented copies (Section 5.1).

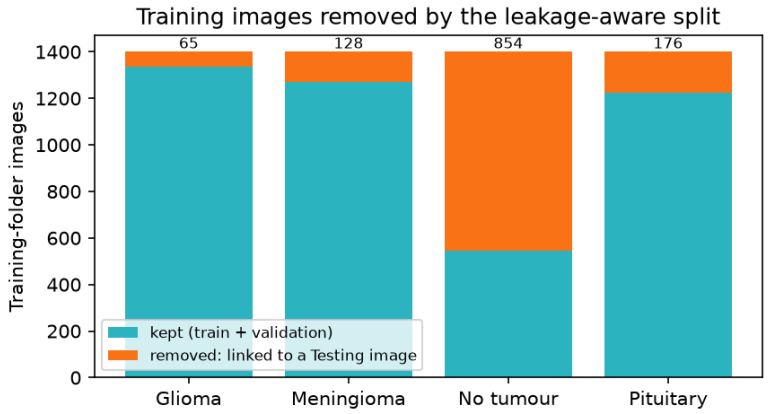


Figure 2: Training-folder images kept and removed by the leakage-aware split, per class.

3.4 Leakage-aware split

We remove from the training pool every Training image whose group contains a Testing image (1,223 images) and every image in a label-conflict group, then assign whole groups to validation (about 15% per class, seeded) and the rest to training. This leaves 3,709 training and 668 validation images, and keeps the official 1,600 Testing images untouched, so the test set is directly comparable with other work that uses the version 2 Testing folder, while no image linked to it under our rule is seen in training. The removal is very uneven (Figure 2): 854 of the 1,400 no-tumour Training images are linked to Testing images, leaving only 454 for training. We also report results on the Testing folder reduced to one image per group (1,342 images).

4 Method

Pre-processing. Every image is converted to greyscale (ITU-R 601 luma), zero-padded to a square to keep its aspect ratio, resized to 224×224 with bilinear filtering, replicated to three channels and normalised with ImageNet statistics. The browser demo implements the same steps.

Model and training. We fine-tune EfficientNet-B0 [25] pretrained on ImageNet (`timm` [26], `efficientnet_b0.ra_in1k`, 4.0 M parameters, dropout 0.2) in PyTorch [17] with AdamW (learning rate 5×10^{-4}, weight decay 10^{-4}), a one-cycle schedule, batch size 32 and 12 epochs. Inverse-frequency class weights compensate for the reduced no-tumour class. Augmentation consists of random resized crops (scale 0.8–1.0, aspect ratio 0.9–1.1), horizontal flips, rotations up to 10° with 5% translation, and brightness/contrast jitter of 0.2. The checkpoint with the best validation macro-F1 is kept (epoch 11). All random generators are seeded (42). Training ran on an 8-core CPU without a GPU in about 115 s per epoch.

Naive comparison. To measure the effect of the split, we train an identical model on the original Training folder with a random, image-level 15% validation split and evaluate both models on the same Testing images.

Calibration and uncertainty. We fit a single temperature on the validation logits by minimising the negative log-likelihood [10]. A prediction is flagged *uncertain* when its calibrated confidence is below a threshold τ, chosen on validation as the smallest value in $\{0.50, 0.51, \ldots, 0.99\}$ for which the retained validation predictions reach at least 99% accuracy. For the leakage-aware model $\tau = 0.69$ (validation coverage 98.7%, retained accuracy 99.1%). The normalised entropy of the calibrated distribution is reported alongside but not thresholded. The same validation set is used for checkpoint selection, temperature fitting and the threshold; it is never used for testing.

Table 1: Accuracy, macro-F1 (95% bootstrap CIs) and calibrated ECE on the Testing folder and subsets. "Has strict dup. in Training" = Testing images with a strict near-duplicate in the original Training folder; "No linked dup." = Testing images with no linked image in it. Macro-F1 is omitted for the class-skewed subsets (the strict-duplicate subset has no pituitary images, so a four-class macro-F1 would count that class as 0; over the three classes present the leakage-aware model's macro-F1 is 0.752).

Model	Test subset	n	Accuracy [95% CI]	Macro-F1 [95% CI]	ECE
Leakage-aware	Full Testing	1600	0.919 [0.905, 0.932]	0.919 [0.905, 0.931]	0.052
Leakage-aware	Testing, one per group	1342	0.919 [0.904, 0.932]	0.912 [0.895, 0.926]	0.051
Leakage-aware	Has strict dup. in Training	441	0.880 [0.850, 0.912]	– (class-skewed subset)	0.092
Leakage-aware	No linked dup. in Training	970	0.925 [0.908, 0.940]	– (class-skewed subset)	0.046
Naive	Full Testing	1600	0.961 [0.951, 0.970]	0.961 [0.950, 0.970]	0.022
Naive	Testing, one per group	1342	0.959 [0.948, 0.969]	0.959 [0.947, 0.969]	0.023
Naive	Has strict dup. in Training	441	0.995 [0.989, 1.000]	– (class-skewed subset)	0.005
Naive	No linked dup. in Training	970	0.940 [0.926, 0.955]	– (class-skewed subset)	0.035

Table 2: Per-class results of the leakage-aware model on the full Testing folder (n = 1,600).

Class	Precision	Recall	F1	ROC-AUC
Glioma	0.968	0.843	0.901	0.968
Meningioma	0.941	0.840	0.888	0.983
No tumour	0.816	0.998	0.898	0.996
Pituitary	0.980	0.995	0.988	1.000
Macro average	–	–	0.919	0.987

Metrics. We report accuracy and macro-F1 with 95% bootstrap confidence intervals (1,000 resamples), per-class precision, recall and F1, the confusion matrix, one-vs-rest ROC-AUC and the expected calibration error (ECE, 15 bins). Resampling is at image level; because some Testing images are near-duplicates of each other, these intervals are slightly optimistic. All numbers in this paper are generated by scripts from a single saved prediction file per model (`results/predictions/`); the same file feeds the demo's statistics.

5 Results

5.1 Classification

Table 1 summarises accuracy on the Testing folder and its subsets, and Table 2 gives per-class results for the leakage-aware model; the confusion matrix and ROC curves are in Figure 3.

The leakage-aware model reaches 91.9% accuracy and macro-F1 0.919 on the full Testing folder, and 91.9% on the one-per-group variant. Errors are not symmetric. Pituitary tumours are almost always recognised (recall 99.5%), and all but one no-tumour image are classified correctly (recall 99.8%), but at a cost in no-tumour precision (81.6%): 42 gliomas and 48 meningiomas are called "no tumour". Glioma recall is the lowest (84.3%) although its precision is 96.8%. Missing a tumour is the clinically more serious error, and in this model it is the most frequent one. Validation accuracy (98.7%) is much higher than test accuracy, which points to a shift between the Training and Testing folders beyond the duplicates we removed (for instance different source collections or acquisition protocols); we could not resolve the source of each image.

Naive versus leakage-aware training. The model trained on the original Training folder scores 96.1% on the same Testing folder (95% CI 95.1%–97.0%), 4.3 points (68 images) above the leakage-aware model. Two effects are mixed in this gap. The naive model sees more data (4,760 instead of 3,709 training images, and 1,190 instead of 454 no-tumour images), which can genuinely help. But Table 3 shows where the gap comes from: on the 105 Testing meningiomas that have a strict

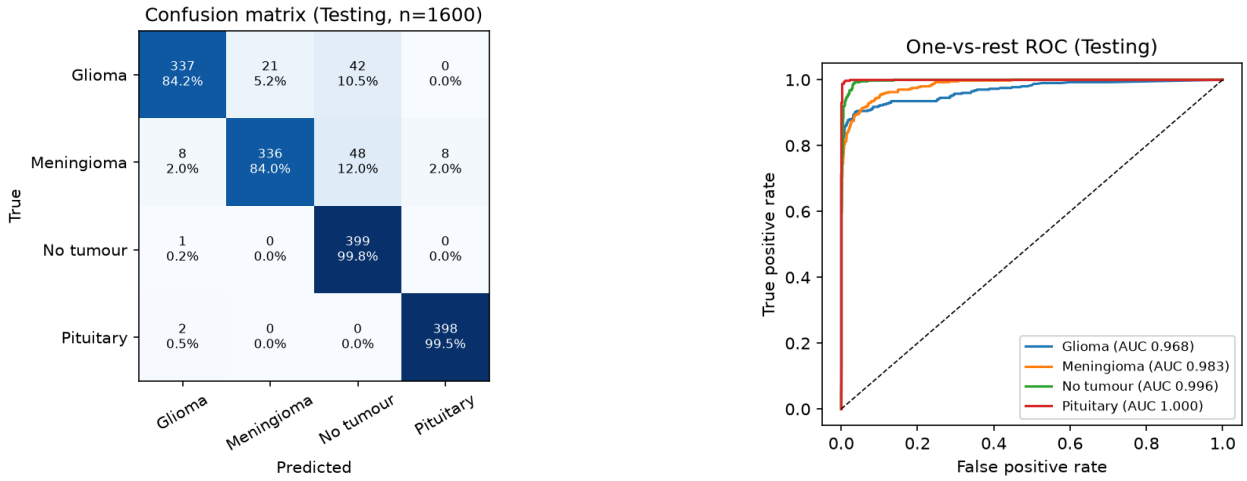


Figure 3: Leakage-aware model on the Testing folder: confusion matrix (counts and row percentages) and one-vs-rest ROC curves.

Table 3: Accuracy (correct/total) per class on Testing images with and without a strict near-duplicate in the original Training folder.

	Strict dup. in Training		No strict dup.	
Class	Leakage-aware	Naive	Leakage-aware	Naive
Glioma	0.778 (7/9)	0.778 (7/9)	0.844 (330/391)	0.867 (339/391)
Meningioma	0.524 (55/105)	1.000 (105/105)	0.953 (281/295)	0.976 (288/295)
No tumour	0.997 (326/327)	1.000 (327/327)	1.000 (73/73)	1.000 (73/73)
Pituitary	–	–	0.995 (398/400)	0.998 (399/400)

near-duplicate in Training, the naive model is right on 105/105 and the leakage-aware model on 55/105, whereas on the meningiomas without one the two are close (288/295 versus 281/295). 50 of the 68-image gap therefore comes from these 105 meningiomas, which suggests that much of the headline difference is recall of near-copies of training images rather than better generalisation. Part of this gap also reflects the leakage-aware model losing all training examples of that sub-population, so the accuracy a leak-free model could reach on it is unknown. No-tumour images are classified almost perfectly by both models whether or not they have a duplicate (one error in total), so the large no-tumour overlap barely changes accuracy here, although it would affect any comparison that relies on that class.

Augmented files. The meningioma class contains files named `*-aug-*` (100 in Training, 103 in Testing), which appear to be augmented copies. Searching all eight flips and 90° rotations, only 26 of them match another image under our "linked" rule (10 only after a flip or rotation), so most are transformed in ways our hash does not detect. Among meningiomas without a strict near-duplicate in Training, the leakage-aware model is correct on 100/102 augmented and 181/193 other files (98.0% versus 93.8%); the naive model, which trained on all augmented Training files, shows no such advantage (97.1% versus 97.9%), so leakage through augmented copies is possible but not demonstrated.

5.2 Calibration and uncertainty

The fitted temperature is 1.91, meaning the raw model is over-confident. On Testing, ECE falls from 0.070 to 0.052 (Figure 4). With $\tau = 0.69$, 3.5% of Testing predictions are flagged uncertain; accuracy is 93.8% on the confident ones and 39.3% on the flagged ones. The flag therefore concentrates errors, but most errors on Testing still carry high confidence, so it cannot be treated as a safety net.

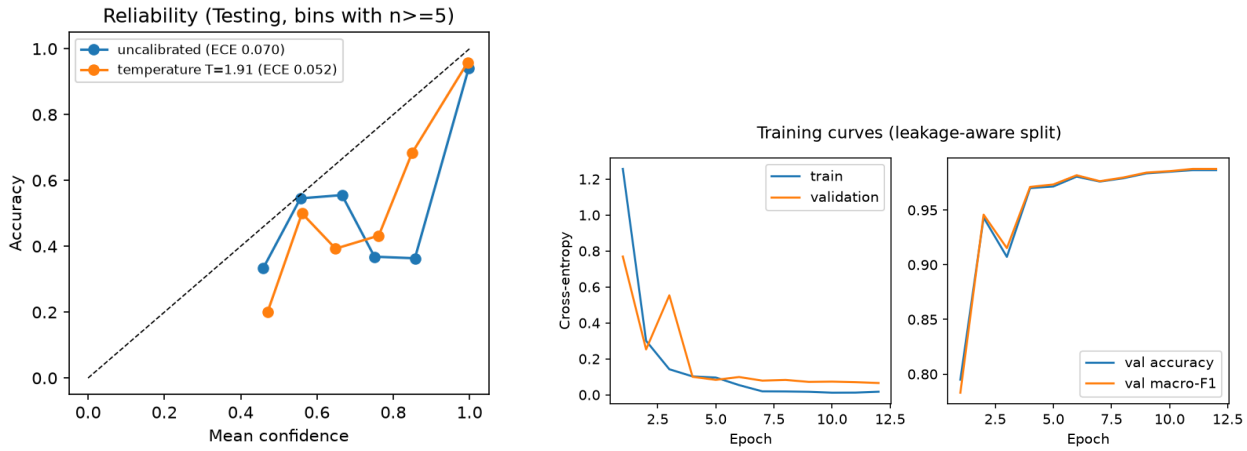


Figure 4: Left: reliability diagram before and after temperature scaling (Testing). Right: training curves of the leakage-aware model.

5.3 Changes from version 1

The first public version of this study (v1, commit `a591cfa`, tag `v1.0`) used only the first two clauses of the linked rule. An internal review found 19 Testing images (18 no tumour, 1 glioma) whose contrast- or sharpening-altered copies at Hamming distance 8–10 were still in our training or validation sets. Version 2 adds the third clause (215 more linked pairs, 1,223 instead of 1,198 Training images removed), retrains the leakage-aware model with identical settings, uses 1,000 bootstrap resamples for every row of Table 1 (v1 used 200 for the two duplicate-defined subsets), corrects several claims and citations, tightens the rule for the demo presets and adds the robustness check of Section 8. Table 4 compares the two versions; the v1 metric files are kept unchanged in `results/v1/`.

Test accuracy falls from 93.4% to 91.9%. This is expected: v2 is the more honest estimate, because the hidden near-copies found by the review are no longer in training (no Testing image now has a training or validation partner with correlation ≥ 0.98 within Hamming distance 10). It is not all due to the rule, however. The training pool (train plus validation) loses only 25 images, 18 no tumour and 7 glioma and no meningioma, although the group-wise train/validation assignment is drawn again; yet meningioma recall changes from 89.8% to 84.0%, mostly on the meningiomas that have a strict near-duplicate in the original Training folder (whose copies were removed in both versions). Part of the difference is therefore run-to-run training variance of a single seed, which our confidence intervals do not capture.

6 Explainability

6.1 Methods

All explanations target the predicted class. **Grad-CAM** weights the final $7 \times 7 \times 1280$ feature map by the spatially averaged gradients; because EfficientNet-B0 ends in global average pooling and a linear layer, these weights equal the classifier weights divided by 49, so Grad-CAM coincides with CAM up to a positive scale. We use this to compute Grad-CAM in ONNX without gradients (maximum difference from autograd Grad-CAM after normalisation 1.6e-05). **Grad-CAM++** uses the closed-form higher-order weights of Chattopadhay et al. [6]. **Integrated Gradients** (Captum [12]) integrates 32 steps from a black image. **Occlusion** slides a black 32×32 patch with stride 16 and records the drop in the class logit. **KernelSHAP** [14] attributes the class logit to about 64 SLIC superpixels [1] using 200 coalitions with a black baseline. Signed maps are clipped (occlusion, SHAP) or taken in absolute value (IG), then min–max normalised.

Table 4: Leakage-aware model, v1 (commit `a591cfa`) versus v2, on the same 1,600 Testing images. All values are read from the saved metric files of each version.

Metric	v1 (a591cfa)	v2
Duplicate link rule	MD5, or Hamming ≤ 6 & $r \geq 0.95$	v1, or Hamming ≤ 10 & $r \geq 0.98$
Linked pairs	5,962	6,177
Training-folder images removed	1,198	1,223
Train / validation images	3,739 / 663	3,709 / 668
Test accuracy (95% CI)	93.4% (92.1%–94.5%)	91.9% (90.5%–93.2%)
Macro-F1	0.933	0.919
Macro ROC-AUC	0.989	0.987
ECE raw $\rightarrow$ calibrated	0.054 $\rightarrow$ 0.036	0.070 $\rightarrow$ 0.052
Temperature T / threshold τ	1.84 / 0.67	1.91 / 0.69
Uncertain rate	3.8%	3.5%
Accuracy confident / uncertain	95.3% / 44.3%	93.8% / 39.3%
Recall glioma	84.0%	84.3%
Recall meningioma	89.8%	84.0%
Recall no tumour	100.0%	99.8%
Recall pituitary	99.8%	99.5%
One-per-group test accuracy	93.1%	91.9%
Naive model accuracy (same test)	96.1%	96.1%
Gap naive − leakage-aware	2.8 pts	4.3 pts
Meningiomas with strict dup.: naive / leakage-aware	105/105 / 74/105	105/105 / 55/105
Bootstrap resamples	1,000 (200 for two subset rows)	1,000 for every row

Table 5: Faithfulness (mean ± s.d. over 100 Testing images), attribution mass on the head region, and Spearman correlation with the trained-model map after randomising the classifier only and after randomising the whole network (12 images).

Method	Deletion AUC ↓	Insertion AUC ↑	Mass on head region	ρ after classifier rand.	ρ after full rand.
Grad-CAM	0.487 ± 0.355	0.841 ± 0.195	79.2%	0.11	-0.07
Grad-CAM++	0.490 ± 0.353	0.842 ± 0.183	84.0%	0.94	0.07
Integrated Gradients	0.283 ± 0.259	0.679 ± 0.252	97.8%	0.94	0.84
Occlusion	0.466 ± 0.333	0.803 ± 0.226	82.7%	0.14	0.03
KernelSHAP (SLIC)	0.400 ± 0.351	0.749 ± 0.254	76.5%	0.24	0.21
Random baseline	0.283 ± 0.149	0.497 ± 0.366	57.9%	–	–

6.2 Evaluation protocol

On 100 Testing images (25 per class, seeded) we compute the *deletion* and *insertion* AUC [18]: pixels are removed (set to black) or revealed (from a Gaussian-blurred copy, kernel 31, $\sigma = 10$) in 32 equal steps in order of decreasing attribution (ties, e.g. zeros after clipping, broken randomly with a fixed seed), and the softmax probability of the target class is integrated over the fraction of pixels. A random attribution serves as baseline. We also measure the share of attribution mass on the bright head region (a crude foreground mask). For the sanity check [2] we cumulatively re-initialise the network from the classifier down to the stem on 12 images and compute the Spearman rank correlation between each method's map from the trained and the randomised model; a method that passes should lose correlation as layers are randomised. If either map is constant the correlation is set to 0 (7 of the 72 Grad-CAM rows, none for the other methods), which slightly favours passing.

6.3 Results

Table 5 and Figure 5 summarise the comparison.

Insertion separates the methods clearly from the random baseline (0.497): Grad-CAM (0.841) and Grad-CAM++ (0.842) rank the pixels that restore the prediction best, followed by occlusion (0.803), KernelSHAP (0.749) and Integrated Gradients (0.679). *Deletion* does not: no method beats the random baseline (0.283); Integrated Gradients only matches it (0.283), and the region-based methods score worse than random. With a black fill, scattered random deletions create out-of-distribution images that the model abandons quickly, while deleting a compact region leaves a natural-looking slice; the deletion metric therefore rewards pixel-scattered maps under this baseline

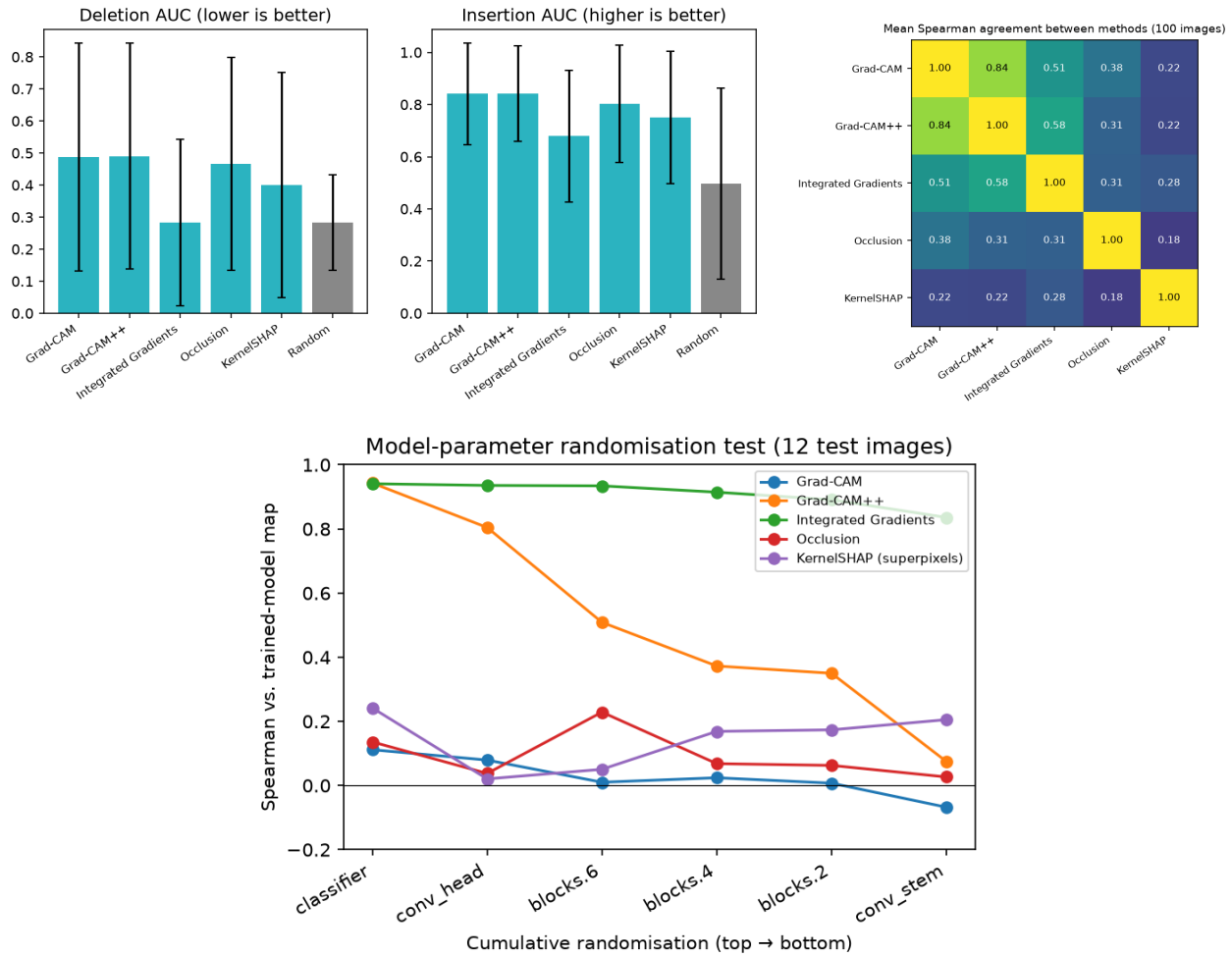
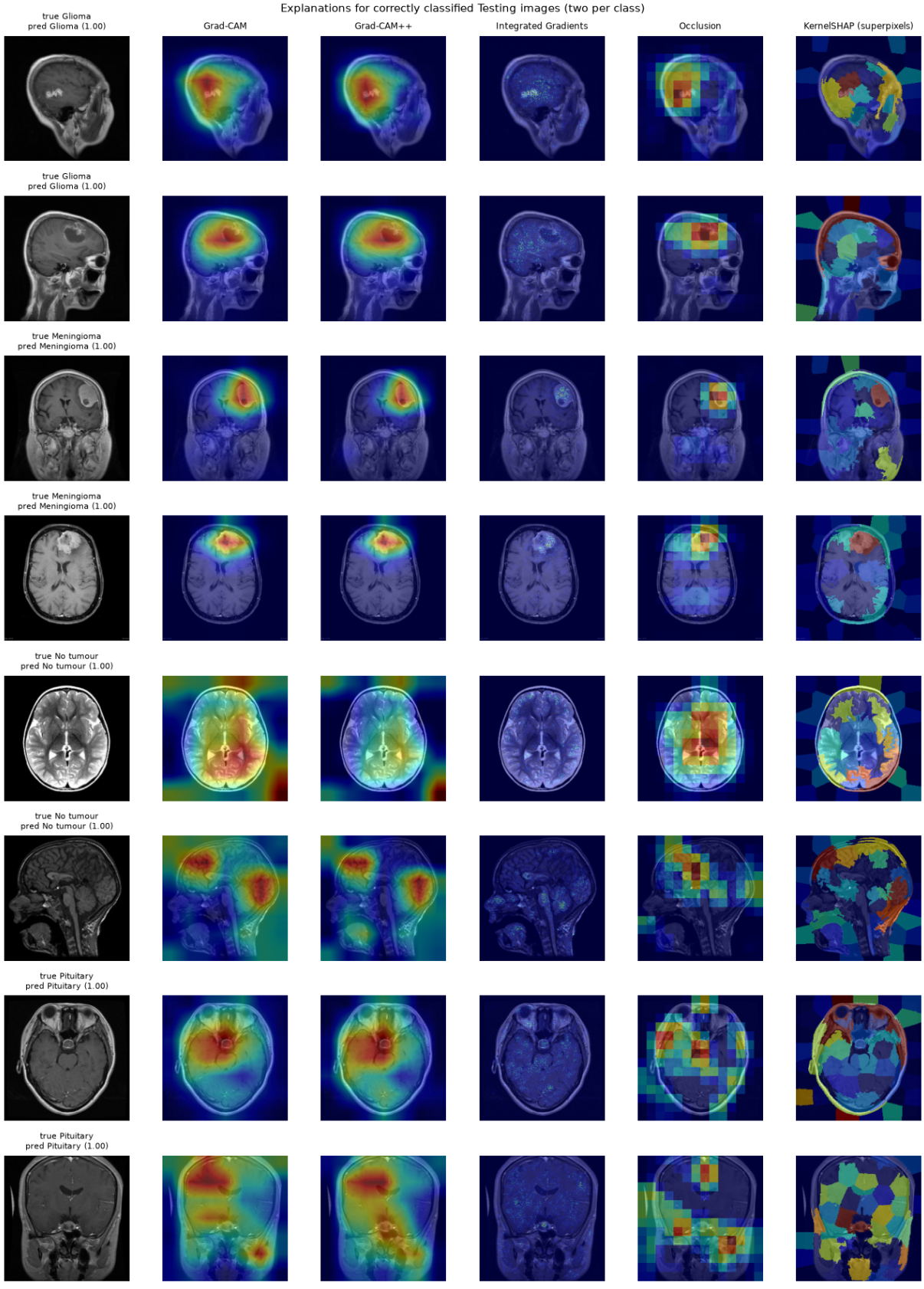


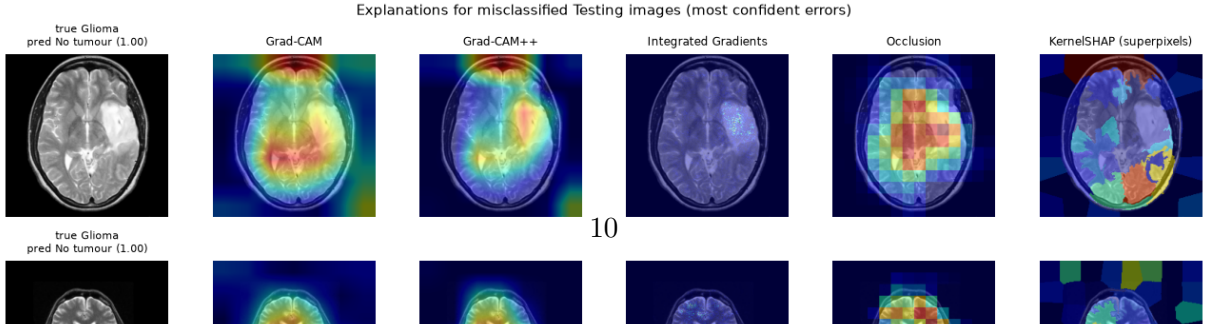
Figure 5: Top: deletion/insertion AUC per method and mean pairwise Spearman agreement. Bottom: cascading model-parameter randomisation.

and should be read with care. Integrated Gradients puts 97.8% of its mass on the head region, the other methods 76–84%.

The sanity check is more decisive. Randomising only the classifier already drops the correlation with the trained-model map to 0.11 for Grad-CAM, 0.14 for occlusion and 0.24 for KernelSHAP, so these maps depend on what the model learned. Grad-CAM++ stays at 0.94 after the classifier is randomised and only falls to 0.07 once the whole network is random, meaning that its maps are largely determined by the feature extractor rather than the class decision. Integrated Gradients fails the test: its maps still correlate at 0.84 with the original after every layer has been re-initialised, the edge-detector behaviour described by Adebayo et al. [2]. Its relatively good deletion score is thus not evidence that it reflects the model. KernelSHAP keeps a small correlation (0.21) even after full randomisation because its superpixels depend on the image, not on the model. Agreement between methods is moderate at best (Spearman 0.84 between Grad-CAM and Grad-CAM++, 0.38 with occlusion, 0.51 with Integrated Gradients and 0.22 with KernelSHAP).

Qualitatively, in the examples in Figure 6, Grad-CAM and occlusion often highlight the lesion in gliomas and meningiomas; for pituitary tumours the maps are less consistent. The most confident errors are mixed: for some gliomas called “no tumour” Grad-CAM falls on the skull rim or away from the visible lesion, while for others it covers the lesion and the prediction is still wrong, so a map that looks plausible does not make the decision correct. We therefore use Grad-CAM, which passes the sanity check and, with Grad-CAM++, has the highest insertion AUC, as the default in the demo, offer occlusion as a model-agnostic cross-check, and present the other maps only as precomputed comparisons.


Explanations for correctly classified Testing images (two per class)
true Glioma
pred Glioma (1.00)
Grad-CAM
Grad-CAM++
Integrated Gradients
Occlusion
KernelSHAP (superpixels)
true Glioma
pred Glioma (1.00)
true Meningioma
pred Meningioma (1.00)
true Meningioma
pred Meningioma (1.00)
true No tumour
pred No tumour (1.00)
true No tumour
pred No tumour (1.00)
true Pituitary
pred Pituitary (1.00)
true Pituitary
pred Pituitary (1.00)


Explanations for misclassified Testing images (most confident errors)
true Glioma
pred No tumour (1.00)
Grad-CAM
Grad-CAM++
Integrated Gradients
Occlusion
KernelSHAP (superpixels)
true Glioma
pred No tumour (1.00)

7 In-browser demonstrator

The leakage-aware model is exported to ONNX (16.0 MB) with two outputs, the logits and the four class activation maps, and runs in the browser with ONNX Runtime Web (WebAssembly); the maximum logit difference to PyTorch on 64 Testing images is 2.4e-04. The page replicates the pre-processing, applies the fitted temperature and shows animated class probabilities, a Grad-CAM overlay with an opacity control and an on-demand occlusion map with the setting evaluated in Section 6 (32-pixel black patch, stride 16, $13 \times 13 = 169$ forward passes). Eight Testing slices are bundled as presets, two per class, drawn with a fixed seed. Preset rule: only use test images with no candidate pair at all at Hamming ≤ 10, whatever the correlation; augmented (`*-aug-*`) files are also excluded. They are bundled with their true labels and with precomputed Grad-CAM++, Integrated Gradients and KernelSHAP maps.

A *Why this prediction* panel is generated from templates filled only with measured quantities: the predicted class and calibrated confidence, the margin over the runner-up class, the uncertainty flag ($\tau = 0.69$) and normalised entropy, the share of Grad-CAM mass on the head region and its dominant image quadrant, the rank agreement between methods for the bundled samples, and that class's test precision and recall with its most common confusion, read from the same statistics file as Table 2. It contains no medical reasoning and is not a diagnosis.

8 Limitations

- **No patient-level split.** The dataset has no patient identifiers. Our groups capture near-identical images and very similar neighbouring slices, but other slices of the same patient may still occur on both sides of the split, so results may remain optimistic.
- **Augmented copies.** Augmented meningioma files in both folders are mostly not detected by our hash and may still leak information, although we found no clear evidence of it (Section 5.1).
- **Thresholds by inspection; pHash can miss copies.** The duplicate thresholds were set by looking at examples, and pHash can still miss some near-copies (for example strongly cropped, rotated or re-contrasted ones), so a small residual overlap may remain. As a robustness check without retraining, a looser rule that also links pairs at Hamming 7–10 with correlation ≥ 0.95 adds 644 pairs and links 84 further Testing images (10 no tumour) to our training or validation images; all of them are classified correctly (100.0%), and excluding them the leakage-aware model's accuracy is 91.4% (n = 1,516) against 91.9% on the full folder. The released candidate pairs allow re-analysis with other thresholds.
- **Source and protocol confounding.** No-tumour images come from a different collection than the tumour classes, so a model can partly learn source characteristics (contrast, resolution, framing) rather than pathology. The gap between validation and test accuracy is consistent with such shifts.
- **Slice-level, single-label task.** Real diagnosis uses volumes, several sequences, clinical context and histopathology; this model sees one 2-D slice and four labels and cannot recognise other conditions.
- **Single run.** We trained one seed per configuration on a CPU; the confidence intervals reflect test-set sampling only, not training variance.
- **Explanations are not localisation.** No tumour masks were used, so we cannot measure whether attributions fall on the lesion; the head-region share is only a sanity proxy.

9 Ethical considerations

The images are public and de-identified, and we redistribute only eight resized samples under the CC BY 4.0 licence with attribution. A tumour classifier can cause harm if its output is mistaken for a diagnosis: it can miss tumours (the most common error here) and give confident wrong answers. The demonstrator therefore states prominently that it is a research and education tool, not a medical device, processes images locally without uploading them, shows calibrated uncertainty, and builds its rationale only from the model's own outputs and measured error rates. Saliency maps can create unwarranted trust [2, 3]; we report their sanity checks and faithfulness rather than presenting them as evidence of correctness.

10 Conclusion

A widely used brain MRI benchmark that reports no train/test overlap at the file level still shares near-duplicate images between its folders, mostly in the no-tumour class. With those removed from training and the training data split by duplicate group, a CPU-trained EfficientNet-B0 reaches 91.9% accuracy on the official Testing folder, with its errors concentrated in tumours called "no tumour". Calibration, an explicit uncertainty threshold, tested explanations and a rationale built from measured statistics make the behaviour of the model easier to inspect, but they do not make it a clinical tool. Patient-level splits, external test sets and lesion annotations are the next requirements for any stronger claim.

Code and data availability. Code, notebooks, split files, hashes, predictions and the demo: `https://github.com/Rishabh-bgp/brain-tumor-mri-classification`. Live demo: `https://rishabh-bgp.github.io/brain-tumor-mri-classification/`. Dataset: Nickparvar [16].

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